Program : **Diploma in Cyber Forensics And Information Security**		
Course Code : **3281**	Course Title: **Fundamentals of Cryptography and Information Security**	
Semester : **3**	Credits: **3**	
Course Category: **Program Core Course**		
Periods per week: **3 (L:3 T:0 P:0)**		Periods per semester: **45**

Course Objectives:

- To understand basics and fundamentals of Information Security and Cryptography.
- To understand the need of security
- To identify various malicious softwares and understand the intrusion detection systems.
- To identify and mitigate DoS attacks and the use of firewalls.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge in mathematics		Mathematics I and II	I, II
Basic knowledge in Computer systems		Introduction to IT Systems Lab	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the basic concepts related to Information security and Cryptography	11	Understanding
CO2	Illustrate the various methods of user authentication	11	Understanding
CO3	Classify various malicious software and explain Intrusion detection systems	9	Understanding
CO4	Understand Denial of service and Firewalls	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Demonstrate the basic concepts related to Information security and Cryptography**		
M1.01	Explain the history of Information Security	1	Understanding
M1.02	Explain the components of an Information system.	1	Understanding
M1.03	Illustrate the need for security, Distinguish Threats and Attacks, Threats and Assets	1	Understanding
M1.04	Explain classical encryption techniques and Symmetric cipher modes	2	Understanding
M1.05	Explain various substitution techniques	2	Understanding
M1.06	Explain various transposition techniques	2	Understanding
M1.07	Demonstrate the use of Steganography in Cryptography	2	Understanding
M1.08	Explain simplified model of symmetric encryption (basic idea only)	1	Understanding
Computer Security : History of Information Security-Components of Information system-Definition –Triad – Confidentiality , Integrity, Availability – Need of Security concepts – relationships – Threats , Attack, Assets Cryptography : Classical Encryption Techniques- Symmetric cipher modes-Substitution techniques - Transposition techniques - Steganography - Symmetric Encryption			

CO2	**Illustrate the various methods of user authentication**		
M2.01	Explain the basic concepts of User Authentication and the means of authentication	1	Understanding
M2.02	Explain Password based Authentication	1	Understanding
M2.03	Explain various Password attack strategies and countermeasures	2	Understanding
M2.04	Describe the use of hashed passwords	1	Understanding
M2.05	Explain various password cracking approaches and also about user password choices	2	Understanding
M2.06	Explain the basic concepts of password File access control	1	Understanding
M2.07	Outline various password selection strategies	1	Understanding
M2.08	Outline various Authentication Methods such as Token based authentication , Biometric Authentication , Remote User Authentication.(Basic idea only)	1	Understanding
M2.09	Relate Access control in computer security, Explain Access Control Principles and various access control policies	2	Understanding
	Series Test -1		
User Authentication: Means of authentication - Password based Authentication - Password attack strategies and countermeasures - hashed passwords - password cracking - user password choices - password File access control - password selection. Authentication Methods: Token based authentication -Biometric Authentication - Remote User Authentication - security issues- Access control: Principles – Relationship among other security functions - access control policies			
CO3	**Classify various malicious software and explain Intrusion detection systems**		
M3.01	Explain various types of Malicious Software - Explain Viruses , The nature of viruses, Virus structure, .	2	Understanding
M3.02	Classify Viruses, Antivirus approaches & Antivirus techniques .	1	Understanding
M3.03	Explain Worms- Outline Worm propagation model - Explain various countermeasures for worm propagation	2	Understanding
M3.04	Outline various classes of intruders and the intruder behavior patterns.	2	Understanding

M3.05	Explain Intrusion Detection System, classification and the requirements of IDS.	2	Understanding
Malicious Software & Intrusion Detection Malicious Software: Different types - Viruses – nature - Virus structure – Classification - Antivirus approaches - Antivirus techniques – Worms - Propagation model - Worm Countermeasures - Intrusion and Detection: Classes of intruders - Intruder behavior patterns - classification - requirements of IDS			
CO4	**Understand Denial of service and Firewalls**		
M4.01	Explain Denial of Service (DoS) and the effect of DoS on Network bandwidth, System resources and Application resources	2	Understanding
M4.02	Interpret the Classic Denial of Service Attacks	2	Understanding
M4.03	Explain the need of firewall and summarize various characteristics of a Firewall	2	Understanding
M4.04	Classify firewalls.	2	Understanding
	Series Test – II	1	
Denial of Service and Firewall Denial of Service: Definition - Effect of DoS on Network bandwidth and System resources -Classic Denial of Service Attacks -: Need - Characteristics - Types of firewalls .			

Text / Reference

T/R	Book Title/Author
T1	Computer Security- Principles and Practice - Author: William Stallings & Lawrie Brown Publisher: Pearson Prentice Hall 2010
T2	Cryptography And Network Security Principles And Practice Fourth Edition, William Stallings, Pearson Education
R1	W. Mao, “Modern Cryptography – Theory and Practice”, Pearson Education.
R2	Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing –Prentice Hall of India.

Online Resources

Sl No	Website Link
1	https://www.tutorialspoint.com/cryptography/pdf/cryptography_quick_guide.pdf
2	https://www.w3schools.com/cybersecurity/cybersecurity_firewalls.php
3	https://www.omnisecu.com/security/index.php